

Fig. 28: Reducing common-mode coupling

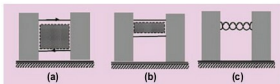


Fig. 29: Reducing differential-mode coupling

to and outputs from adjustable speed motor drives, inputs to and outputs from SMPS or DC-DC converters, cables associated with electrical welders or RF equipment (like induction welders, wood gluers, microwave equipment and diathermy apparatus) and cables to DC motors, to RF transmitting antennae and to untransmitted inductive loads.

Classes 5 and 6. Class 5 cables are those that carry medium voltage, while class 6 are those that carry high voltage.

Cable segregation

During their run in a system, cables of different classes have to be properly segregated, that is, separated by certain minimum distance. This separation distance not only depends on class but also on cable length. Fig. 27 shows recommended minimum segregation distance for different classes of cables based upon a run of 30 metres, and assuming that the cables run in close proximity to the protective earth conductor that can be a wire, a metal tray or mounting plate that forms a local RF reference.

Accordingly, class 1 and class 2 cables must be separated by a minimum distance of 150mm (ditto for class 3 and class 4). While class 1 and class 3 and class 2 and class 4 should be separated by at least 450mm, and class 1 and class 4 by at least 600mm, separation between class 4 and class 5 and class 5 and

class 6 should be 150mm. Now, these distances are for a cable run of less than 30 metres. For longer cables, separation distance is multiplied by the length and divided by 30.

When cables run within a product, these must be physically segregated depending upon their class at all times, and long parallel runs should be avoided. If that is not possible, these should be more than 150mm apart; cable classes—as far as possible—must not cross. If at all these have to cross, these should do so at right angles. And cables should run as close as possible to their local RF reference or to cables carrying return signals. Cables of the same class, when routed through the same tray or duct, should not be twisted around one another.

Reducing common-mode coupling

When cables are routed away from the ground plane or earth, a ground loop is formed as shown in Fig. 28. This loop acts as an antenna for stray fields causing common-mode EMI to be coupled into the system. By the same token, the loop can also cause radiated emission.

Coupling or radiation is directly proportional to the area of the loop. In order to reduce it, area of the loop must be reduced. This is done by routing the cable as close as possible to the ground reference, which, in case of control panels, is the mounting plate.

Reducing differential-mode coupling

This type of coupling occurs when send and return cables are routed away from each other. As shown in Fig. 29(a), a loop is formed between send and return paths, causing differential-mode coupling and radiation

proportional to the area of the loop.

This can be reduced by reducing the area of the loop, that is, by routing send and return cables close to each other. Further reduction can be achieved by twisting the cables around one another, whereby the loop area is reduced dramatically.

Cable routing in electronic control panel

Fig. 30 shows cable trunking for a control panel used for controlling a typical machine. All components should be mounted on an unpainted mounting plate for proper grounding connections for the modules as well as their cables. The three-phase mains comes from a very hostile environment, which is a characteristic of shop floors, and carries signals that can be strongly interfering. Such cables are therefore classified as class 4.

It is essential that these signals are filtered immediately after entering the cabinet, which is normally achieved by a suitable line filter. The filtered cable is classified as class 3, which then goes to the SMPS and the drive. Output of the SMPS is again classified as class 4 since it can carry strong interference. Filter at PLC I/O converts this into a class 3 cable, which then goes to the PLC. I/O lines of PLC carry control signals that can be slightly sensitive and, hence, are classified as class 2. Some class 2 lines from PLC go as control cables to the drive.

As a rule, shielded cables must be used for such lines with the shield exposed and bonded to the mounting plate through 360-degree saddle clamps. Moreover, proper distance has to be maintained between these and nearby class 3 and class 4 cables. In Fig. 30, this distance is shown as 300mm and 450mm, respectively; although it may be less considering small cable lengths (remember the 300mm distance is for lines in excess of 30m).

Some I/O cables of the PLC go to